5 Fig. 5.1 shows a kitchen tap that supplies instant boiling water.

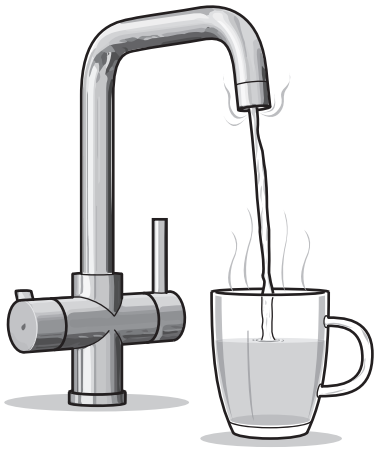


Fig. 5.1

Cold water passes over an electric immersion heater inside the tap.

The boiling point of water is 100 °C.

(a) State what is meant by boiling point.

.....

..... [2]

(b) The immersion heater is powered by the mains at a voltage of 230 V. When the tap is opened, the heater switches on and the current in the heater is 13 A.

(i) Calculate the thermal energy produced by the heater in 60 s.

thermal energy = ... [2]

(ii) The specific heat capacity of water is 4200 J / (kg °C). The cold water that enters the tap is at 22 °C.

Calculate the rate at which water at its boiling point emerges from the tap.

rate = ... [4]

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(c) The metal tap is earthed and there is a fuse in the cable that connects the heater to the mains.

1. Explain how the earth wire protects the user.

.....

.....

2. Explain how the fuse protects the circuit.

.....

.....

[3]

[Total: 11]